Cost-Effective Sensing Structure For Characterization of Physical Interaction Between User's Upper Limbs and Smart Walkers

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***Abstract*—This paper presents the design and a preliminary functional validation of a low-cost force-sensing structure to measure the physical interaction between a user's upper limbs and a smart walker. Configured as a Hand Support (HS), the system was statically characterized and calibrated using loads up to 20 kg, applied in 0.5 kg increments up to 10 kg, and 5 kg steps thereafter. Subsequently, its functionality was assessed against a commercial reference load cell acting as a Forearm Support (FS). Acting as a physical human-robot interface, the proposed structure provided the input signals for an admittance controller, enabling natural and proportional control of a smart walker, UFES WalkerXR, velocity and direction. Static calibration demonstrated reliable measurement performance, yielding a full-scale error of less than 0.5 percent and high linearity with an R-squared value greater than 0.999. Furthermore, a functional validation case study involving a single participant navigating straight-line, Z-shape, and lemniscate paths revealed a dynamic behavior analogous to the commercial sensor. The system achieved low Root Mean Square Error (RMSE) values during trajectory tracking (e.g., straight-line: 0.0599 m for FS vs. 0.0709 m for HS; lemniscate: 0.0941 m for FS vs. 0.1032 m for HS). Additionally, the interaction data highlighted distinct kinematic and biomechanical differences between the support modes, such as variations in user distance and load distribution. Ultimately, the results indicate that the developed sensing structure provides consistent static measurements and suitable functional performance, facilitating the physical human-robot interaction required for smart walker control.**

***Index Terms*—Force Sensor, Low-cost, Physical Human Robot Interfaces, Sensing Structure, Smart Walker.**

I. Introduction

Recent reports from the World Health Organization (WHO) highlight a steady increase in global population aging, with projections estimating 2.1 billion older adults by 2050 [1]. The WHO has classified this demographic shift as a public health priority because the physiological frailty inherent to aging directly predisposes individuals to a state of disability [2], [3]. However, advanced age is not the sole trigger for this condition, as severe neurological events such as cerebrovascular accidents can also induce frailty and disability in patients of any age [3], [4]. Regardless of its etiology, disability adversely impacts the physical, mental, and social domains of the patient, severely compromising their autonomy and overall quality of life [4]. In response to these challenges, the WHO advocates for the strengthening of healthcare systems and the development of assistive technologies to support activities of daily living (ADLs) [5]. Beyond daily assistance, these devices can also serve as rehabilitation tools [6]. Depending on specific clinical needs, these technologies aim to support the recovery of the functional, cognitive, and psychosocial capacities of affected individuals [7], [8].

Among these technologies, smart walkers represent a versatile platform to achieve these therapeutic goals. These robotic platforms are instrumented with sensors, actuators, and control algorithms that enable functionalities such as partial body weight support, gait monitoring, and real-time feedback [9], [10]. This level of instrumentation allows them to provide cognitive assistance for clinical manifestations like spatial disorientation through adapted interfaces [11], [12], while simultaneously offering physical support that goes beyond merely improving stability and safety during walking [1], [13] to actively facilitate post-stroke rehabilitation processes [6].

In smart walkers, force sensors are widely employed for quantifying user forces to infer movement intention [14] and ensuring a safe, fluid physical human-robot interaction (pHRI) [6]. Consequently, commercial sensors are widely used across various frameworks, due to their established reliability

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[15], including admittance control for intuitive navigation [6], [16], virtual visual feedback [17], fall prevention [1], [18], and machine learning-based clinical gait analysis [19], [20]. However, the high cost of these commercial force sensing systems represents a barrier to their real-world implementation, restricting the practical impact of these technologies in clinical and community settings, particularly in low- and middle-income countries [5], [21].

To address this accessibility challenge, several studies have explored alternative sensory configurations. For instance, Leite et al. [22], and Wang et al. [23] instrumented smart walkers handlebars with resistance-based force sensors to infer users' movement intentions from grip and unloading force patterns. In a related approach, Ding et al. [24] integrated four modified handlebars with similar resistance-based sensors into a forearm-support walker, two in the standard grip configuration and two in an alternative hand-support arrangement, enabling the measurement of grip, unloading, and pushing forces under different support modes. Moving beyond resistance-based force sensors, other researchers have implemented low-cost load cells to quantify unloading patterns in forearm-support four-wheeled walkers [25].

Similar to the systems previously described, most low-cost structures rely on a single force component, measuring either grip force or unloading force. Furthermore, literature indicates that these alternatives present limitations in tasks requiring precision, such as force interactions to capture user intention [15]. Consequently, a gap exists for hand-grip designs that quantify both unloading and pushing forces concurrently. To address this gap, the present work proposes a low-cost force-sensing structure, designed as a Hand Support (HS), to quantify the concurrent interaction forces exerted by the user's upper limbs in smart walker applications. The proposed structure was statically calibrated and subsequently validated in a functional case study, in which it was integrated into the UFES WalkerXR smart walker and its functionality was compared against a commercial reference sensor in a Forearm Support (FS) configuration.

II. MATERIALS AND METHODS

A. UFES WalkerXR Integration

In this study, the UFES WalkerXR [27] was utilized as the experimental platform. This smart walker is actuated by two DC motors (Imobras, Brazil) and achieves state estimation through a combination of two incremental encoders (Autonics Corporation, South Korea) and an inertial measurement unit (Xsens, The Netherlands). Additionally, a LiDAR sensor (SLAMTEC, China) provides real-time tracking of the user's lower limbs. The platform also integrates a commercial reference system consisting of forearm supports (FS) instrumented with biaxial load cells (ProCal, Brazil; estimated cost of $5,500–$6,000 USD), as well as the proposed low-cost Hand Support (HS) sensing structure (estimated cost of $104–$134 USD), which is detailed in Section II-B. Both configurations measure the unloading forces (UF) and pushing forces (PF) exerted by the user. These force readings serve as the primary input for the walker's embedded admittance controller [28], which modulates the kinematics of the robotic platform to assist navigation according to the user's movement intention.

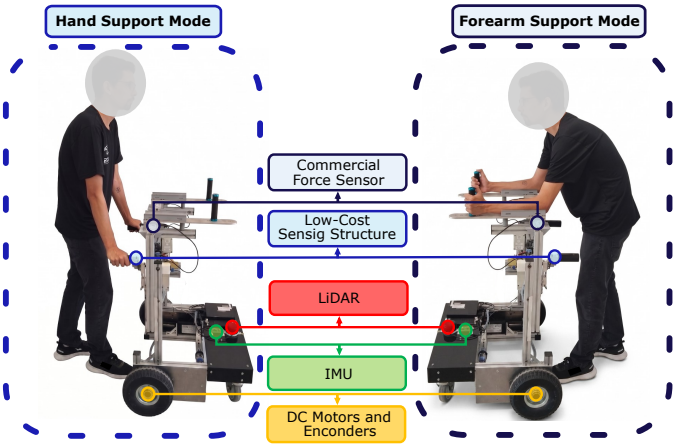


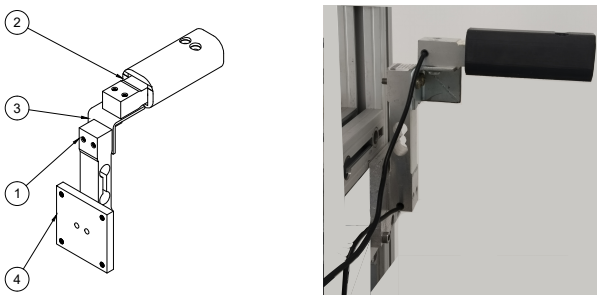
Fig. 1: UFES WalkerXR platform highlighting the commercial reference sensors, the proposed low-cost Hand Support, and the integrated navigation hardware.

B. Hand Sensor Structure

1) Hardware Design: The sensory system comprises two 40 kg load cells (IndiPeso LTDA, Brazil), used for measuring vertical unloading force (UF) and anterior-posterior push force (PF). Their differential output signals were digitized by HX711 24-bit ADC modules (Avia Semiconductor, China) powered at 5 V to improve dynamic range and reduce quantization error. A bidirectional logic level converter (Nexperia, Netherlands) adapted these 5 V signals to the 3.3 V logic of an STM32F411CEU6 microcontroller (STMicroelectronics, France). Data acquisition was managed by this microcontroller at a sampling frequency of 10 Hz. Rather than applying explicit digital filters, the system relies on this low sampling rate and the walker's high mechanical inertia to naturally attenuate high-frequency gait vibrations.

The mechanical structure comprises three coupled components configured to direct interaction forces to their respective sensors independently (see Fig. 2a). The outer housing (part 2) functions as an ergonomic handgrip, compliant with the ISO 11199-2 diameter specifications [26], and encloses the UF load cell (part 1). To provide a customizable interface, this grip was fabricated via Fused Deposition Modeling 3D printing using acrylonitrile butadiene styrene (ABS). To ensure structural rigidity and prevent parasitic deformations during load transfer, the assembly incorporates a custom-machined galvanized steel L-bracket (part 3) that serves as the fixed cantilever end for the UF sensor while axially transmitting the pushing motion to the PF sensor. Finally, a machined aluminum base (part 4) anchors the PF sensor and clamps the entire assembly to the UFES WalkerXR frame.

2) Characterization and Static Calibration: Prior to their mechanical adaptation to the smart walker, the load cells were initially characterized under controlled static conditions



(a) Exploded CAD view (b) Physical implementation

Fig. 2: Proposed force-sensing structure showing (a) the component assembly including the load cells (1), ABS grip (2), galvanized steel L-bracket (3), and aluminum walker coupling (4), and (b) the integrated system mounted on the walker's frame.

to determine their raw response. To establish characterization curves and assess hysteresis effects, a bi-directional loading protocol was executed along each measurement axis. Incremental loads were applied in 0.5 kg steps from 0 kg to 10 kg to capture fine variations within the primary interaction range, followed by 5 kg steps up to 20 kg. This sequence was then identically reversed for the unloading phase, utilizing a 20-second dwell time per step in both directions to ensure signal stabilization. During data processing, transient regions were discarded, and the mean value of the steady-state response was computed for each load level. Finally, a linear regression model was fitted to these stable readings to extract the sensitivity (slope) and offset (intercept) characterizing each measurement axis.

During the subsequent calibration phase, these derived parameters were implemented in the microcontroller firmware to convert raw digital readings into physical units (kgf), and the static zero-load output was algorithmically subtracted to establish a zero baseline. A metrological evaluation was then conducted to verify this calibration. The corrected physical outputs were compared against the reference loads, targeting an nominal post-calibration sensitivity of 1.0 kgf and an offset of 0.0 kgf. Sensor linearity was assessed via the coefficient of determination (R^2), while measurement error was quantified using the Mean Absolute Error (MAE) in kilograms. This error was further normalized to the 20 kg measurement range and expressed as a percentage of the Full-Scale Range (%FSR). Hysteresis was calculated as the maximum output divergence between the loading and unloading cycles at identical load levels, also reported as a %FSR. Finally, to verify measurement independence, cross-axis interference was evaluated; the maximum parasitic load induced on the orthogonal physical axes (unloaded axes) during primary axis loading was recorded and reported as a %FSR to quantify structural coupling effects.

C. Experimental Protocol

The experimental tests involved a 26-year-old healthy volunteer with a body mass of 56.9 kg and a height of 1.50 m. To assess the sensing interfaces during various navigational scenarios, the volunteer navigated three pre-established trajectories using the smart walker. Path 1 consisted on a straight 5.5 m path to observe continuous linear forward force. Path 2 consisted on a Z-shaped trajectory featuring a 2.7 m initial segment, a 90° right turn, a 3.0 m intermediate segment, a 90° left turn, and a 4.0 m final segment, designed to evaluate sharp turning maneuvers.Finally, path 3 was represented by a Bernoulli lemniscate curve ($\alpha = 2.3$ m) to assess continuous steering and directional changes.

During the validation trials, the smart walker was operated under two distinct pHRI modes, as illustrated in Fig. 1. The first mode utilized a reference interface consisting of forearm supports (FS) instrumented with commercial biaxial load cells (ProCal, Brazil). The second mode employed the low-cost hand-support (HS) sensing structure presented in this study, which is detailed in Section II-B. Both sensory configurations measure the UF and PF applied by the user. These force readings are subsequently input into the walker's admittance controller [28] to modulate its kinematics. To ensure that any differences in human-robot interaction were attributable to the sensory interfaces rather than algorithmic variations, the admittance controller parameters, specifically the virtual mass (m) and damping (d) for translational (v) and rotational (w) motion, were kept constant across all experiments ($m_v = 1.0$, $m_w = 0.1$, $d_v = 40.0$, $d_w = 8.0$).

The volunteer performed a single trial per path for each sensing interface. To accommodate the volunteer's anthropometry, the height of the supports was adjusted to 48% [28] and 61.8% [29] the volunteer's total height for the FS and HS configurations, respectively.

Throughout the execution of the paths, the system recorded odometry, interaction forces, and LiDAR-based user distance measurements. For data analysis, the raw UF and PF signals from the left and right sensors were processed independently. To characterize the variability of force application, the mean and standard deviation (SD) of the forces for each axis (UF and PF) were computed for each sensor.

Furthermore, odometry data facilitated the calculation of the Root Mean Square Error (RMSE). This metric, computed as the root mean square of the minimum Euclidean distances between each recorded point of the user's trajectory and the ideal reference path, was selected to evaluate spatial deviation. Additionally, the mean linear velocity and its SD were calculated to evaluate the overall speed and continuity of the navigation under each support type. Finally, LiDAR data allowed for the monitoring of the distance between the user's feet and the walker's frame, utilizing the mean and SD to assess user proximity. The results for these metrics were grouped by interaction mode (FS versus HS) to establish a preliminary functional validation of the hardware's capability to drive the walker.

III. RESULTS AND ANALYSIS

A. Static Calibration

Table I presents the sensitivity (slope) and zero-load offset (intercept) parameters that define the static response curve

for each measurement axis. The coefficient of determination ($R^2 \geq 0.99$) across all axes confirms that the proposed mechanical assembly preserves the inherent linearity of the load cells, preventing any structural deformations or non-linear compliance that could degrade sensor performance. Furthermore, the varying non-zero offset values represent the static pre-load exerted by the mass of the mechanical components resting on each cell. To compensate for this structural tare, a dynamic zeroing routine was implemented in the microcontroller firmware. During startup, this baseline offset is compensated for computing the average of the first ten digital readings and applying the derived sensitivity to establish a zero-force reference before initiating human-robot interaction.

TABLE I: Static characterization parameters for the sensing structures.

Axis	Sensitivity (kgf/ADC)	Offset (kgf)	R^2
Left Push	9.72×10^{-6}	0.68	1.00
Left Weight	-9.71×10^{-6}	-1.09	1.00
Right Push	1.31×10^{-5}	-0.41	1.00
Right Weight	-8.98×10^{-6}	0.39	1.00

The metrological evaluation of the characterization and calibration process yields metrics that confirm the linearity of the sensing system. The coefficient of determination (R^2) remained consistent at 1.00 for all evaluated load cells across the three trials. Furthermore, the sensitivity values presented in Table II approximate with an theoretical ratio of 1.00. This demonstrates that the firmware adjustments in the microcontroller established a proportional relationship between the applied mechanical load and the digital output. Note that the initial characterization sensitivity (kgf/ADC) reflects the raw ADC response, whereas this calibrated sensitivity ($\approx$1 kgf) verifies the expected linear behavior of the digital readings once converted into physical weight.

The absolute errors recorded during the trials indicate the system's accuracy and repeatability, yielding an average Mean Absolute Error (MAE) of 0.03 kgf across both axes of each sensor, differentiating the Unloading Force (UF) and Push Force (PF) measured by the left (UF-L, PF-L) and right (UF-R, PF-R) sensing structures. Coupled with standard deviations remaining below 0.01 kgf, these results highlight a consistent calibration protocol that is further corroborated by the full-scale percentage error (%FSR) reported in Table II. Beyond absolute error, the hysteresis percentages detailed in the same table demonstrate consistent sensor precision throughout the loading and unloading phases. Although mechanical variances within the assembly cause the UF-R cell to exhibit higher hysteresis than the other units, this deviation remains within acceptable limits and does not compromise measurement integrity for the current application.

Regarding baseline stability, the dynamic zeroing process maintained low zero-load offsets. Specifically, the left sensors exhibited mean offsets of -0.01 kgf (UF-L) and 0.04 kgf (PF-L), while the right sensors recorded 0.00 kgf (UF-R) and 0.05 kgf (PF-R), with all associated standard deviations remaining below 0.01 kgf. Furthermore, an analysis of cross-axis interference indicates low structural coupling within the operational range. The absolute maximum induced loads (crosstalk) were 0.20 ± 0.01 kgf for UF-L and 0.35 ± 0.02 kgf for PF-L, alongside 0.33 ± 0.06 kgf for UF-R and 0.08 ± 0.01 kgf for PF-R. As shown in Table II, this interference remains below 5%FSR across all axes, providing sufficient structural decoupling for this application and bypassing the need for computational decoupling algorithms. Notably, the PF-R sensor registered the lowest cross-axis effect; while minor mechanical variations in the assembly can influence such measurements, this particular variance increases isolation without negatively impacting overall metrological performance.

TABLE II: Main Metrological Parameters (Mean $\pm$ SD, n=3).

Sensor	Sensitivity (kgf)	Error (%FSR)	Hysteresis (%FSR)	Crosstalk (%FSR)
UF-L	1.04 $\pm$ 0.00	0.17 $\pm$ 0.00	0.02 $\pm$ 0.00	1.01 $\pm$ 0.04
PF-L	1.03 $\pm$ 0.00	0.17 $\pm$ 0.00	0.03 $\pm$ 0.00	1.73 $\pm$ 0.08
UF-R	1.01 $\pm$ 0.00	0.16 $\pm$ 0.00	0.11 $\pm$ 0.16	1.66 $\pm$ 0.28
PF-R	1.00 $\pm$ 0.00	0.17 $\pm$ 0.00	0.02 $\pm$ 0.00	0.39 $\pm$ 0.03

B. Functional Validation

In Path 1 (Figs. 3a and 3b), PF curves exhibit similar magnitudes for both FS and HS, reflecting the balanced one-to-one limb ratio required for straight-line advancement. Initially in Fig. 3b (0 s to 3 s), the volunteer applied a higher load to the right handle, but the load subsequently balanced across both handles. Additionally, the force signals exhibit oscillatory patterns that can be associated with the volunteer's gait cadence, a characteristic also observed across all evaluated paths (see Fig. 3). Furthermore, these oscillatory patterns allow the observation of distinct phase behaviors where the HS UF signals exhibit an in-phase pattern of simultaneous fluctuation (see Fig. 3b), whereas FS displays an out-of-phase pattern (see Fig. 3a). This FS behavior indicates alternating support during the gait cycle, likely loading the right UF during the right-foot stance and the left UF during the left-foot stance.

Regarding Path 2 (Figs. 3c and 3d), the out-of-phase UF pattern in Fig. 3c becomes more pronounced during turning events. These events highlight the PF differentials driving the right turn (FS: 12 s to 20 s; HS: 15 s to 25 s) and left turn (FS: 24 s to 33 s; HS: 32 s to 40 s). Initial HS UF signals (see Fig. 3d, up to 20 s) exhibit fluctuations indicative of initial grip adjustments, but subsequent normalization reflects load stabilization, consistent with Path 3 (Figs. 3e and 3f). Furthermore, a phase shift in HS UF forces near the start and end of turns (e.g., see Fig. 3d at t = 22 s) likely reflects the biomechanical adjustments required to redirect the device.

Finally, Path 3 (Figs. 3e and 3f) confirms the previously observed out-of-phase UF behavior. Simultaneously, PF patterns map the 2.5 repetitions of the lemniscate trajectory curves. Overall, while absolute magnitudes are not strictly comparable between support modes due to biomechanical differences related to the distinct postural configuration of each support mode, the sensing structure effectively captures



Fig. 3: Comparison of interaction forces for Forearm (FS, left) and Hand Support (HS, right) modalities across Path 1 (a, b), Path 2 (c, d), and Path 3 (e, f).

the force differentials that drive movement via the admittance controller.

The overall performance of the participant is presented in Table III. The mean and standard deviation of the interaction forces directly reflect the user's posture and physical coupling with the smart walker. The mean force values reveal a more symmetrical load distribution in the HS mode compared to the FS mode, with UF values exhibiting comparable magnitudes between the left and right sides. Furthermore, consistent with the temporal signal analysis, while the magnitude of the PF differs due to the biomechanical nature of the supports, their dynamic behavior is analogous; this consistency is reflected in the uniformity of the left and right PF measurements within each support condition. Regarding force variability, the standard deviations generally tended to be lower in the FS compared to the HS, indicating greater load stability when FS mode is utilized.

This difference in load stability could corresponds to distinct kinematic behaviors. The measured user distance was consistently lower in the FS mode, indicating that the user's center of mass remained closer to the center of the smart walker. This proximity minimizes relative displacement between the user and the device, as evidenced by the lower standard deviation of the distance metrics, thereby facilitating more controlled locomotion, highlighting how body disposition alters the magnitude and distribution of the interaction forces. Consequently, the admittance controller facilitated higher average walking speeds and more consistent motion with the FS. Specifically, velocity variability in the FS mode exceeded that of the HS mode only during the straight-line path. In contrast, the HS mode exhibited greater variability during turning trajectories, suggesting that the closer center of mass in the FS mode facilitate rotational maneuvers. Finally, the Root Mean Square Error (RMSE) of the trajectory tracking indicates that the FS mode yields more accurate path execution. Nevertheless, tracking errors for both support modes remained within acceptable

TABLE III: Kinematic and Interaction Forces Results across Support Modes (FS and HS)

Metric	Mode	Path 1 (Straight)	Path 2 (Z-Shape)	Path 3 (Lemniscate)
UF - Left (kgf)	FS	4.42 ± 1.18	6.49 ± 1.46	7.14 ± 1.28
	HS	6.30 ± 1.71	4.21 ± 1.64	6.23 ± 1.37
UF - Right (kgf)	FS	3.61 ± 0.91	5.70 ± 1.63	5.41 ± 2.00
	HS	6.04 ± 1.82	4.05 ± 1.51	6.19 ± 1.32
PF - Left (kgf)	FS	0.85 ± 0.37	0.77 ± 0.82	0.71 ± 1.25
	HS	0.65 ± 0.39	0.84 ± 1.01	0.71 ± 1.33
PF - Right (kgf)	FS	0.83 ± 0.34	0.75 ± 0.82	1.11 ± 1.23
	HS	0.64 ± 0.23	0.74 ± 0.74	1.11 ± 1.23
User Distance (m)	FS	0.68 ± 0.03	0.66 ± 0.03	0.65 ± 0.04
	HS	0.76 ± 0.03	0.82 ± 0.06	0.78 ± 0.09
Velocity (m/s)	FS	0.21 ± 0.04	0.19 ± 0.04	0.22 ± 0.03
	HS	0.17 ± 0.03	0.16 ± 0.07	0.22 ± 0.06
RMSE (m)	FS	0.06	0.07	0.09
	HS	0.07	0.10	0.10

limits, demonstrating that the proposed HS configuration is a viable interface for controlling the smart walker.

IV. CONCLUSIONS

A force sensing structure based on low-cost components was developed, undergoing static validation and preliminary dynamic assessment via a single case study. The system exhibited a Mean Absolute Error (MAE) of a few hundredths of a kilogram with strictly limited standard deviation. Crosstalk remained below 2% FSR, a negligible value. These outcomes, coupled with high linearity ($R^2 > 0.99$), low full-scale error ($< 0.5\%$ FSR), and minimal hysteresis ($< 0.5\%$ FSR), demonstrate robust force measurements under static conditions.

Furthermore, the dynamic trials demonstrated the system's capability to operate the UFES WalkerXR, capturing PF patterns consistent with the reference sensor. While providing a stable initial baseline, the differences observed in load patterns and force magnitudes warrant subsequent investigations. Given the limitation of a single-participant case study, these biomechanical and kinematic differences represent preliminary findings and cannot be generalized without subsequent kinetic and kinematic investigations involving a broader population.

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